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310406

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310406

B. Tech. III - Sem. (Main) Exam., February - 2021

Computer Science & Engineering

3CS4-06 Object Oriented Programming

Common for CS/ IT

Time: 3 Hours **Maximum Marks: 120**

Instructions to Candidates:

Part – A: *Short answer questions (up to 25 words) 10 × 2 marks = 20 marks. All ten questions are compulsory.*

Part – B: *Analytical/Problem Solving questions (up to 100 words) 5 × 8 marks = 40 marks. Candidates have to answer five questions out of seven.*

Part – C: *Descriptive/Analytical/Problem Solving questions 4 × 15 marks = 60 marks. Candidates have to answer four questions out of five.*

Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of following supporting materials is permitted during examination. (Mentioned in form No. 205)

1. NIL 2. NIL

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PART - A

Q.1 What is abstract class? [2]

Q.2 What is the purpose of defining a destructor function? [2]

Q.3 What is dynamic memory allocation? [2]

Q.4 What is function overriding? [2]

Q.5 Define 'this' Pointer. [2]

Q.6 What is virtual class? [2]

Q.7 Name all Access modifiers. [2]

Q.8 What is exception? [2]

Q.9 What is Polymorphism? [2]

Q.10 What is template? [2]

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PART – B

Q.1 List out difference between Procedure oriented programming and Object oriented programming. [8]

Q.2 Write a C++ program to swap two numbers using pointer. [8]

Q.3 Explain function overloading in C++ with an example. [8]

Q.4 Write a C++ program to overload + operator to add two complex numbers. [8]

Q.5 Explain multiple inheritance in C++ with example. [8]

Q.6 Define virtual function. Explain the mechanism of virtual function. [8]

Q.7 What is the use of new and delete operator in C++? Explain with example. [8]

PART – C

Q.1 What are the different types of constructors in C++? Illustrate with an example. [15]

Q.2 Explain types of inheritance in C++ with suitable example. [15]

Q.3 What is an exception? Explain exception – handling mechanism with suitable example. [15]

Q.4 Explain run time polymorphism, its advantages and how it is implemented in C++. [15]

Q.5 What is file? Write various file handling functions in C++. How file handling is done in C++? Explain with example. [15]

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